

Legislative Analysis



MICHIGAN PLANE COORDINATE SYSTEM

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<http://www.house.mi.gov/hfa>

House Bill 5577 as introduced
Sponsor: Rep. John R. Roth
Committee: Regulatory Reform
Complete to 3-4-26

Analysis available at
<http://www.legislature.mi.gov>

SUMMARY:

House Bill 5577 would amend 1964 PA 9, which the bill would rename the State Plane Coordinates Act.

The bill would change the system of plane coordinates used by the National Oceanic and Atmospheric Administration/National Geodetic Survey (NOAA/NGS) for defining and stating the positions of points on or near the surface of the Earth in Michigan from the Michigan Coordinate System (MCS) of 1983 to the *Michigan Plane Coordinate System*.¹ Positions in the new system would still be expressed as x- and y-coordinates and would still need to conform with the values published by NOAA/NGS.

Michigan Plane Coordinate System would mean the system of plane coordinates, which has been established by NOAA/NGS, based on the National Spatial Reference System (NSRS), and known as the SPCS, for defining and stating the geographic positions or locations of points in Michigan.

Zones

The 1983 MCS divides the state in a north, central, and south zone and allows the position of all points a tract of land extending from one zone to another be referred to either of the zones, as long as the zone used is identified in the description of the tract.

The bill would eliminate the existing zones and instead define *zone* to mean the geographic areas encompassed by closed contiguous county boundaries or submerged lands, or both.

In addition, when a tract of land crosses zone boundaries, any relevant zone could still be used to refer to all points of the tract but the zone used would have to be included in the tract's *metadata*, in addition to the description, as currently required.

Metadata would mean data describing other data. It would include at least the geodetic reference system used, applicable epoch, statement of relative accuracy, and date of observation.

Requirements for recording

The act currently requires that coordinates purporting to define the position of a point or land boundary using the 1983 MCS can only be presented to be recorded by a land surveyor if they express an estimate, expressed as a standard deviation, of the positional tolerance of the position being recorded. The recording document is also required to contain a description of

¹ The act also refers to the 1927 MCS but required its use be discontinued beginning in 1990.

the nearest first or second order horizontal geodetic control monument from which the coordinates being recorded were determined and the method of survey for that determination.

Under the bill, the estimate of positional tolerance would have to be presented at a 95% confidence interval. In addition, the requirement to include information about the nearest horizontal geodetic control monuments would be replaced by a requirement to include information about the metadata and horizontal geodetic control monuments from which the coordinates being recorded were determined and the method of survey for that determination.

Other provisions and repeals

The bill would add a requirement that the official geodetic reference system to which geodetic coordinates, including latitude, longitude, ellipsoid height, orthometric height, or dynamic height, are referenced in Michigan be defined by the NSRS.

The bill would repeal various sections that are specific to the MCS of 1927 and 1983.

MCL 54.231 et seq. (amended) and MCL 54.232, 54.235, and 54.235a (repealed)

FISCAL IMPACT:

House Bill 5577 is unlikely to affect costs or revenues for the Department of Natural Resources or the Department of Environment, Great Lakes, and Energy.

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■ This analysis was prepared by nonpartisan House Fiscal Agency staff for use by House members in their deliberations and does not constitute an official statement of legislative intent.